

Lecture 1: Some Basic Results in Probability and Statistics

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Outline

- 1 Summation and Product Operators
- 2 Random variables
- 3 Normal probability distribution and related distributions
- 4 Inferences of the population mean
- 5 Comparisons of two population means
- 6 Inferences about population variance

Summation Operator

$$\sum_{i=1}^n Y_i = Y_1 + \dots + Y_n$$

Properties:

- $\sum_{i=1}^n k = nk$
- $\sum_{i=1}^n (Y_i + Z_i) = \sum_{i=1}^n Y_i + \sum_{i=1}^n Z_i$
- $\sum_{i=1}^n (a + bY_i) = na + b\sum_{i=1}^n Y_i$, where a, c are constants

Summation and Product Operators

Double summation operator:

$$\begin{aligned}\sum_{i=1}^n \sum_{j=1}^m Y_{ij} &= \sum_{i=1}^n (Y_{i1} + \dots + Y_{im}) \\ &= Y_{11} + \dots + Y_{1m} + \dots + Y_{n1} + \dots + Y_{nm}\end{aligned}$$

Properties: $\sum_{i=1}^n \sum_{j=1}^m Y_{ij} = \sum_{j=1}^m \sum_{i=1}^n Y_{ij}$.

Product operator:

$$\prod_{i=1}^n Y_i = Y_1 \cdot Y_2 \dots Y_n$$

Expected Value

Let the random variable Y with the list of the outcomes $Y_1, \dots, Y_k$ with probabilities of occurring are given by:

$$f(Y_s) = P(Y = Y_s), s = 1, \dots, k$$

Expected value of Y , $E\{Y\}$, is defined by:

$$E\{Y\} = \sum_{s=1}^k Y_s f(Y_s).$$

If Y a continuous random variable with density function $f(Y)$:

$$E\{Y\} = \int_{-\infty}^{\infty} Y f(Y) dY.$$

Properties: $E\{a\} = a$; $E\{cY\} = cE\{Y\}$; $E\{a + cY\} = a + cE\{Y\}$, where a, c are constants.

Variance

The variance of the random variable Y , $\text{var}\{Y\}$, is defined as follows:

$$\begin{aligned}\text{var}\{Y\} &= E\{(Y - E\{Y\})^2\} \\ &= E\{Y^2\} - (E\{Y\})^2\end{aligned}$$

If Y a continuous random variable,

$$\text{var}\{Y\} = \int_{-\infty}^{\infty} (Y - E\{Y\})^2 f(Y) dY.$$

Properties: $\text{var}\{a + Y\} = \text{var}\{Y\}$, $\text{var}\{cY\} = c^2 \text{var}\{Y\}$, where a, c are constants.

Joint, Marginal, and Conditional Probability Distributions

Joint probability function for the two random variables Y and Z :

$$g(Y_s, Z_t) = P(Y = Y_s \cap Z = Z_t), s = 1, \dots, k, t = 1, \dots, m$$

Marginal probability function of Y is:

$$f(Y_s) = \sum_{t=1}^m g(Y_s, Z_t), s = 1, \dots, k$$

Conditional probability function:

$$g(Y_s|Z_t) = \frac{g(Y_s, Z_t)}{h(Z_t)}, s = 1, \dots, k$$

(showing on diagrams, an example with a table with two columns
“isFemale” and “> 1.7m”)

The covariance of Y and Z , $\text{cov}\{Y, Z\}$, is defined by:

$$\begin{aligned}\text{cov}\{Y, Z\} &= E\{(Y - E\{Y\})(Z - E\{Z\})\} \\ &= E\{YZ\} - (E\{Y\})(E\{Z\})\end{aligned}$$

Properties:

- $\text{cov}\{a_1 + c_1Y, a_2 + c_2Z\} = c_1c_2\text{cov}\{Y, Z\}$
- $\text{cov}\{Y, Y\} = \text{var}\{Y\}$

Standardized Form, Coefficient of Correlation

The standardized form of a random variable Y is defined as:

$$Y' = \frac{Y - E\{Y\}}{\sqrt{\text{var}\{Y\}}}$$

Coefficient of correlation of two RVs Y and Z :

$$\begin{aligned}\rho\{Y, Z\} &= \text{cov}\{Y', Z'\} \\ &= \frac{\text{cov}\{Y, Z\}}{\sqrt{\text{var}\{Y\}}\sqrt{\text{var}\{Z\}}}\end{aligned}$$

Properties

- $-1 \leq \rho\{Y, Z\} \leq 1$
- When $\rho\{Y, Z\} = 0$ ($\text{cov}\{Y, Z\} = 0$ equivalently), Y and Z are said to be uncorrelated.

Independent Random Variables

Two RVs Y and Z are independent if and only if:

$$g(Y_s, Z_t) = f(Y_s)h(Z_t), s = 1, \dots, k, t = 1, \dots, m$$

If Y and Z are independent RVs, then $\rho\{Y, Z\} = 0$ ($\text{cov}\{Y, Z\} = 0$ equivalently).

Central Limit Theorem

If $Y_1, \dots, Y_n$ are independent random observations from a population with probability function $f(Y)$ for which $\text{var}\{Y\}$ is finite, the sample mean $\bar{Y}$:

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

is approximately normally distributed when the sample size n is reasonably large, with mean $E\{Y\}$ and variance $\text{var}\{Y\}/n$.

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Normal probability distribution

- Density function of $N(\mu, \sigma^2)$:

$$f(Y) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{Y - \mu}{\sigma} \right)^2 \right],$$

where μ and σ are two parameters of the distribution.

- $E\{Y\} = \mu$, $\text{var}\{Y\} = \sigma^2$.
- If Y is a normal random variable, the random variable $z = \frac{Y - \mu}{\sigma}$ follows *standard normal distribution*, $N(0, 1)$.

Standard normal random variable

- Let $P\{z \leq z(A)\} = A$. The cumulative probabilities A for percentiles $z(A)$ is provided by Table B.1, Kutner's book.
- E.g., when $z(A) = 2.00$, $A = 0.9772$.
- Because the normal distribution is symmetrical about 0, when $z(A) = -2.00$, $A = 1 - 0.9772 = 0.0228$.
- `scipy.stats.norm(Python)`: `import scipy.stats as stats;`
`stats.norm.ppf(0.9772)`

Some properties of normal distribution

- **Linear function of normal random variable:** If Y is a normal random variable, the transformed variable $Y' = a + cY$ (a and c are constants) is normally distributed with mean $a + cE\{Y\}$ and variance $c^2\text{var}\{Y\}$.
- **Linear combination of independent normal random variables:** When $Y_1, \dots, Y_n$ are independent normal random variables, the linear combination $a_1Y_1 + a_2Y_2 + \dots + a_nY_n$ is normally distributed with mean $\sum_i^n a_i E\{Y_i\}$ and variance $\sum_i^n a_i^2 \text{var}\{Y_i\}$.

χ^2 distribution

- Let $z_1, \dots, z_v$ be v independent **standard normal** random variables. We then define a chi-square random variable as follows:

$$\chi^2(v) = z_1^2 + z_2^2 + \dots + z_v^2$$

The χ^2 distribution has parameter v , which is called the **degrees of freedom** (df).

- Mean: $E\{\chi^2(v)\} = v$
- Percentiles of various χ^2 distributions are given by Table B.3, Kutner's book, $P\{\chi^2(v) \leq \chi^2(A; v)\} = A$.
- `scipy.stats.chi2` (Python)

t Distribution

- t random variable is defined as follows:

$$t(\nu) = \frac{z}{\left[\frac{\chi^2(\nu)}{\nu} \right]^{1/2}},$$

where $z \sim N(0, 1)$ and $\chi^2 \sim \chi^2(\nu)$ and are independent.

- t distribution has one parameter, the degrees of freedom ν .
- Mean $E\{t(\nu)\} = 0$
- Percentile $P\{t(\nu) \leq t(A; \nu)\} = A$ is given by Table B.2.
- *E.g.*, suppose $\nu = 10$. The 90th percentile of the t distribution with 10 degrees of freedom is $t(0.90; 10) = 1.372$.
- Because the t distribution is symmetrical about 0, we have $t(0.10; 10) = -1.372$.
- `scipy.stats.t` (Python)

F distribution

- Let $\chi^2(v_1)$ and $\chi^2(v_2)$ be two independent χ^2 random variables. F random variable is defined as follows:

$$F(v_1, v_2) = \frac{\chi^2(v_1)}{v_1} : \frac{\chi^2(v_2)}{v_2}.$$

- Percentiles of various F distributions is given by Table B.4
 $P\{F(v_1, v_2) \leq F(A; v_1, v_2)\} = A.$
- E.g.*, suppose $v_1 = 2, v_2 = 3$. The 90th percentile of the F distribution with 2 and 3 degrees of freedom, respectively, in the numerator and denominator is $F(0.90; 2, 3) = 5.46$.
- $F(A; v_1, v_2) = \frac{1}{F(1-A; v_2, v_1)}$
- `scipy.stats.f` (Python)

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Inferences of the population mean - Normal population

Given a *random sample* of n observations $Y_1, \dots, Y_n$ from a normal population with mean μ and standard deviation σ .

- The *sample mean* and *sample standard deviation* are:

$$\bar{Y} = \frac{\sum_i Y_i}{n}$$
$$s = \left[\frac{\sum_i (Y_i - \bar{Y})^2}{n - 1} \right]^{1/2}$$

- Estimated standard deviation of the sampling distribution of $\bar{Y}$ is:

$$s\{\bar{Y}\} = \frac{s}{\sqrt{n}}$$

$\frac{\bar{Y} - \mu}{s\{\bar{Y}\}}$ is distributed as t with $n - 1$ degrees of freedom when the random sample is from a normal population.

Interval estimation

The confidence limits for μ with confidence coefficient $1 - \alpha$ are obtained by means of

$$\bar{Y} \pm t(1 - \alpha/2; n - 1)s\{\bar{Y}\}.$$

Example 1: Obtain a 95 percent confidence interval for μ when $n = 10$, $\bar{Y} = 20$, $s = 4$.

- We have $s\{\bar{Y}\} = s/\sqrt{n} = 1.265$. Hence, the confident limit with confidence coefficient 95% ($\alpha = 0.05$) is

$$\bar{Y} \pm t(0.975; 9) * 1.265 = 20 \pm 2.262 * 1.265.$$

- Therefore, the 95 percent confidence interval for of population mean μ is $17.14 \leq \mu \leq 22.86$

Hypothesis testing

One-sided and two-sided tests concerning the population mean μ are constructed by means of $\frac{\bar{Y} - \mu}{s\{\bar{Y}\}}$, based on the test statistic:

$$t^* = \frac{\bar{Y} - \mu_0}{s\{\bar{Y}\}}.$$

	Alternatives	Decision rule
(a) two-sided	$H_0 : \mu = \mu_0$ $H_a : \mu \neq \mu_0$	if $ t^* \leq t(1 - \alpha/2; n - 1)$, conclude H_0 if $ t^* > t(1 - \alpha/2; n - 1)$, conclude H_a
(b) left-sided	$H_0 : \mu \geq \mu_0$ $H_a : \mu < \mu_0$	if $t^* \geq t(\alpha; n - 1)$, conclude H_0 if $t^* < t(\alpha; n - 1)$, conclude H_a
(c) right-sided	$H_0 : \mu \leq \mu_0$ $H_a : \mu > \mu_0$	if $t^* \leq t(1 - \alpha; n - 1)$, conclude H_0 if $t^* > t(1 - \alpha; n - 1)$, conclude H_a

Example 2

Choose between the alternatives: $H_0 : \mu \leq 20$ or $H_a : \mu > 20$ when α is to be controlled at 0.05 and the sample has $n = 15$, $\bar{Y} = 24$, $s = 6$.

- We have

$$s\{\bar{Y}\} = \frac{s}{\sqrt{n}} = \frac{6}{\sqrt{15}} = 1.549$$

$$t(1 - \alpha; n - 1) = t(0.95; 14) = 1.761$$

- Since

$$t^* = \frac{\bar{Y} - \mu_0}{s\{\bar{Y}\}} = \frac{24 - 20}{1.549} = 2.58 > 1.761,$$

we reject H_0 in favour of H_a .

Example 3

Choose between the alternatives: $H_0 : \mu = 10$ or $H_a : \mu \neq 10$ when α is to be controlled at 0.02 and the sample has $n = 25$, $\bar{Y} = 5.7$, $s = 8$.

- We have

$$s\{\bar{Y}\} = \frac{s}{\sqrt{n}} = \frac{8}{\sqrt{25}} = 1.6$$

$$t(1 - \alpha/2; n - 1) = t(0.99; 24) = 2.492$$

- Since $|t^*| = \left| \frac{\bar{Y} - \mu_0}{s\{\bar{Y}\}} \right| = \left| \frac{5.7 - 10}{1.6} \right| = 2.69 > 2.492$, we conclude H_a .

p -value for sample outcome

- The p -value for a sample outcome is the probability that the sample outcome could have been more extreme than the observed one when $\mu = \mu_0$.
Or: The P-value is the smallest significance level α that leads us to reject the null hypothesis.
- A test can be carried out by comparing the p -value with the specified α risk. If the p -value equals or greater than the specified α , H_0 is concluded. If the p -value is less than α , H_a is concluded.

p -value for sample outcome - Example 5

In Example 3, $t^* = -2.69$.

- the one-sided P-value, $P\{t(24) < -2.69\}$, is between 0.005 and 0.0075.
- One-sided P-value is 0.0064.
- Because the test is two-sided and the t distribution is symmetrical, the two-sided P-value is twice the one-sided value, or $2(0.0064) = 0.013$.
- Hence, for $\alpha = 0.02$, we conclude H_a .

Relation between tests and confidence intervals

- There is a direct relation between tests and confidence intervals. For example, the two-sided confidence limits $\bar{Y} \pm t(1 - \alpha/2; n - 1)s\{\bar{Y}\}$ can be used for testing: $H_0 : \mu = \mu_0$ and $H_a : \mu \neq \mu_0$.
- If μ_0 is contained within the $1 - \alpha$ confidence interval, then the two-sided decision rule with level of significance α , will lead to conclusion H_0 , and vice versa.
- Similar correspondences between one-sided confidence intervals and one-sided decision rules.

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Independent samples

Assume that two normal populations have means μ_1 and μ_2 and have the same standard deviation σ . There are two independent samples for each of the two populations: sample 1 is $Y_1, \dots, Y_{n_1}$ and sample 2 is $Z_1, \dots, Z_{n_2}$.

- Estimators of the two population means are the sample means:

$$\bar{Y} = \frac{\sum_i Y_i}{n_1}, \quad \bar{Z} = \frac{\sum_i Z_i}{n_2}$$

- An estimator of $\mu_1 - \mu_2$ is $\bar{Y} - \bar{Z}$
- An estimator of the common variance σ^2 is:

$$s^2 = \frac{\sum_i (Y_i - \bar{Y})^2 + \sum_i (Z_i - \bar{Z})^2}{n_1 + n_2 - 2}$$

- An estimator of $\text{var}\{\bar{Y} - \bar{Z}\}$, the variance of the sampling distribution of $\bar{Y} - \bar{Z}$, is:

$$s^2\{\bar{Y} - \bar{Z}\} = s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$

Comparisons of two population means: CI

$\frac{(\bar{Y} - \bar{Z}) - (\mu_1 - \mu_2)}{s\{\bar{Y} - \bar{Z}\}}$ is distributed as t with $n_1 + n_2 - 2$ degrees of freedom when the two independent samples come from normal populations with the **same standard deviation**.

Interval estimation. The confidence limits for $\mu_1 - \mu_2$ with confidence coefficient $1 - \alpha$ are obtained:

$$(\bar{Y} - \bar{Z}) \pm t(1 - \alpha/2; n_1 + n_2 - 2)s\{\bar{Y} - \bar{Z}\}$$

Comparisons of two population means - Example 6

- Obtain a 95 percent confidence interval ($\alpha = 0.05$) for $\mu_1 - \mu_2$ when:
 - $n_1 = 10$, $\bar{Y} = 14$, $\sum(Y_i - \bar{Y})^2 = 105$
 - $n_2 = 20$, $\bar{Z} = 8$, $\sum(Z_i - \bar{Z})^2 = 224$
- We have

$$s^2 = \frac{\sum_i(Y_i - \bar{Y})^2 + \sum_i(Z_i - \bar{Z})^2}{n_1 + n_2 - 2} = \frac{105 + 224}{10 + 20 - 2} = 11.75$$

$$s^2\{\bar{Y} - \bar{Z}\} = s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right) = 11.75\left(\frac{1}{10} + \frac{1}{20}\right) = 1.7625$$

$$s\{\bar{Y} - \bar{Z}\} = \sqrt{1.7625} = 1.328$$

$$t(1 - \alpha/2; n_1 + n_2 - 2) = t(0.975; 28) = 2.048$$

- Hence, the 95 percent confidence interval for $\mu_1 - \mu_2$ is:

$$(14 - 8) - 2.048(1.328) \leq \mu_1 - \mu_2 \leq (14 - 8) + 2.048(1.328)$$

$$3.3 \leq \mu_1 - \mu_2 \leq 8.7$$

Comparisons of two population means - Hypothesis testing

One-sided and two-sided tests concerning $\mu_1 - \mu_2$ are constructed from t distribution of $\frac{(\bar{Y} - \bar{Z}) - (\mu_1 - \mu_2)}{s\{\bar{Y} - \bar{Z}\}}$ with $df = n_1 + n_2 - 2$. There are three possible cases, based on the test statistic:

	Alternatives	Decision rule
(a) two-sided	$H_0 : \mu_1 = \mu_2$ $H_a : \mu \neq \mu_0$	if $ t^* \leq t(1 - \alpha/2; n_1 + n_2 - 2)$, conclude H_0 if $ t^* > t(1 - \alpha/2; n_1 + n_2 - 2)$, conclude H_a
(b) left-sided	$H_0 : \mu_1 \geq \mu_2$ $H_a : \mu_1 < \mu_2$	if $t^* \geq t(\alpha; n_1 + n_2 - 2)$, conclude H_0 if $t^* < t(\alpha; n_1 + n_2 - 2)$, conclude H_a
(c) right-sided	$H_0 : \mu_1 \leq \mu_2$ $H_a : \mu_1 > \mu_2$	if $t^* \leq t(1 - \alpha; n_1 + n_2 - 2)$, conclude H_0 if $t^* > t(1 - \alpha; n_1 + n_2 - 2)$, conclude H_a

Test - Example 7

- Choose between the alternatives: $H_0 : \mu_1 = \mu_2$ or $H_a : \mu \neq \mu_0$ when α is to be controlled at 0.10 and the data are those of Example 6.
- $t(0.95; 28) = 1.701$, using the decision rule set (a).
- Since $|t^*| = |(14 - 8)/1.328| = |4.52| = 4.52 > 1.701$, we conclude H_a .
- The one-sided P-value here is the probability $P\{t(28) > 4.52\}$. From Table of t distribution, the exact one-sided P-value is 0.00005. Hence, the two-sided P-value is 0.0001. For $\alpha = 0.10$, the appropriate conclusion therefore is H_a .

Paired Observations

When the observations in the two samples are paired (e.g., attitude scores Y_i and Z_i for the i th sample employee before and after a year's experience on the job), we use the differences: $W_i = Y_i - Z_i$, $i = 1, \dots, n$.

The random variable $\frac{\bar{W} - (\mu_1 - \mu_2)}{s\{\bar{W}\}}$ is distributed as t with $n - 1$ degrees of freedom when the differences W_i can be considered to be observations from a normal population and

$$\bar{W} = \frac{\sum_i W_i}{n}, \quad s^2\{\bar{W}\} = \left(\frac{\sum_i (W_i - \bar{W})^2}{n - 1} / n \right)$$

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Inferences about population variance

When sampling from a normal population, the following holds for the sample variance s^2 :

$\frac{(n-1)s^2}{\sigma^2}$ is distributed as χ^2 with $n - 1$ degrees of freedom when the random sample is from a normal population.

Interval estimation: The lower confidence limit L and the upper confidence limit U in a confidence interval for the population variance σ^2 with confidence coefficient $1 - \alpha$ are

$$L = \frac{(n-1)s^2}{\chi^2(1 - \alpha/2; n-1)}; \quad U = \frac{(n-1)s^2}{\chi^2(\alpha/2; n-1)}$$

Inferences about population variance - Example 8

- Obtain a 98 percent confidence interval for σ^2 , using the data of Example 1 ($n = 10, s = 4$). We have:
 $s^2 = 16$; $\chi^2(0.01; 9) = 2.09$; $\chi^2(0.99; 9) = 21.67$
- The 98 percent confidence interval for σ^2 therefore is:

$$6.6 = \frac{9(16)}{21.67} \leq \sigma^2 \leq \frac{9(16)}{2.09} = 68.9$$

Inferences about population variance - Hypothesis testing

One-sided and two-sided tests concerning the population variance σ^2 are constructed by the distribution of $\frac{(n-1)s^2}{\sigma^2}$.

Alternatives	Decision rule
$H_0 : \sigma^2 = \sigma_0^2$ $H_a : \sigma^2 \neq \sigma_0^2$	if $\chi^2(\alpha/2; n-1) \leq \frac{(n-1)s^2}{\sigma_0^2} \leq \chi^2(1-\alpha/2; n-1)$, conclude H_0 Otherwise, conclude H_a .
$H_0 : \sigma^2 \geq \sigma_0^2$ $H_a : \sigma^2 < \sigma_0^2$	if $\frac{(n-1)s^2}{\sigma_0^2} \geq \chi^2(\alpha; n-1)$, conclude H_0 if $\frac{(n-1)s^2}{\sigma_0^2} < \chi^2(\alpha; n-1)$, conclude H_a
$H_0 : \sigma^2 \leq \sigma_0^2$ $H_a : \sigma^2 > \sigma_0^2$	if $\frac{(n-1)s^2}{\sigma_0^2} < \chi^2(1-\alpha; n-1)$, conclude H_0 if $\frac{(n-1)s^2}{\sigma_0^2} \geq \chi^2(1-\alpha; n-1)$, conclude H_a

Summary

- Normal probability distribution and related distributions
- Confident interval
- Inferences of the population mean
- Comparisons of two population means
- Inferences about population variance

- 1 Appendix A, Applied Linear Regression Models, Michael H. Kutner *et al.*, 2004
- 2 STAT451, Penstate. <https://online.stat.psu.edu/stat415/>